Liver Ultrasound AI

Automated Quantitative Ultrasound Feature Analysis & Screening Report

Scan File: 24.jpg

Analyzed Date: 2026-09-20 01:55

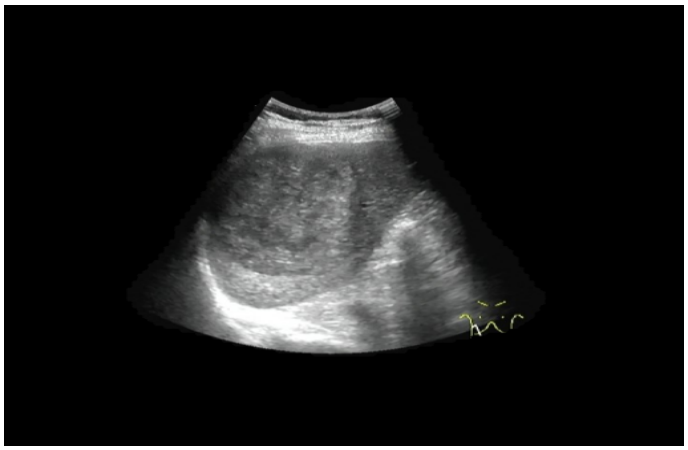
Model Engine: Tuned SVM (RBF kernel)

Clinical Status: Automated Screening Complete

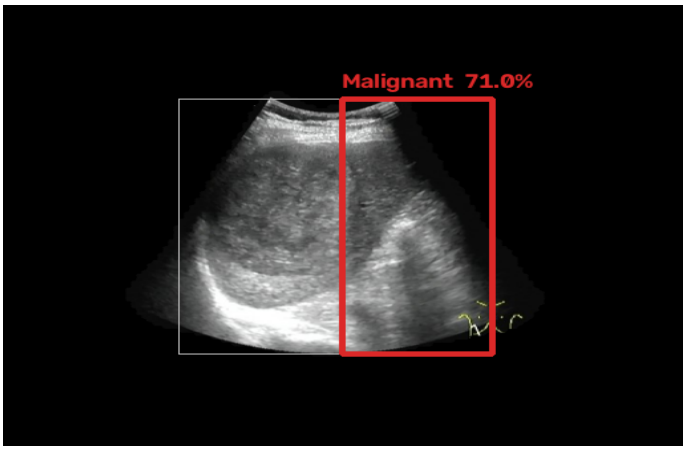
VERDICT: MALIGNANT

Confidence: **71.0%** | Region: **Detected**

Class	Probability	Clinical Interpretation
Malignant	71.0%	Suspicious focal hypoechoic lesion / potential malignancy
Benign	23.5%	Diffuse changes / fatty liver / benign lesion
Normal	5.5%	Healthy parenchymal echotexture, non-fatty liver



Preprocessed Input Scan



Annotated Detection Output

Quantitative Ultrasound Feature Analysis

Comparison of 20 IVP features against Normal liver reference baseline values

#	Extracted Quantitative Feature	Scan Value	Normal Reference	Deviation
1	Liver mean intensity	97.97	80.08 ± 9.5289	+17.8890
2	Liver intensity variation	69.12	61.57 ± 4.9928	+7.5498
3	Liver GLCM contrast	1.0850	0.9150 ± 0.3309	+0.1700
4	Liver GLCM energy	0.1874	0.2921 ± 0.0612	-0.1047
5	Liver GLCM homogeneity	0.7900	0.8475 ± 0.0347	-0.0575
6	Liver edge density	5.51%	4.32% ± 1.16%	+0.0119
7	Candidate detected	Yes	Yes ± No	+0.1200
8	Candidate relative area	27.76%	6.05% ± 7.86%	+0.2171
9	Candidate mean intensity	42.26	30.38 ± 23.87	+11.8837
10	Candidate intensity variation	34.09	11.12 ± 7.0843	+22.9715
11	Candidate/liver intensity ratio	0.4314	0.3740 ± 0.2885	+0.0574
12	Candidate/liver contrast	0.5686	0.5060 ± 0.3147	+0.0626
13	Candidate GLCM contrast	1.0890	0.6147 ± 0.6719	+0.4743
14	Candidate GLCM energy	0.4426	0.5207 ± 0.2802	-0.0781
15	Candidate GLCM homogeneity	0.9069	0.7748 ± 0.3008	+0.1321
16	Candidate circularity	0.1501	0.5121 ± 0.2552	-0.3620
17	Candidate extent	57.98%	52.28% ± 20.80%	+0.0570
18	Candidate solidity	63.00%	81.96% ± 30.88%	-0.1896
19	Candidate aspect ratio	0.4768	1.2337 ± 1.5240	-0.7569
20	Candidate edge density	3.79%	5.38% ± 8.26%	-0.0159

CLINICAL DISCLAIMER: This automated quantitative report is generated for decision support and research evaluation purposes only. It is not an automated medical diagnosis. Final diagnostic evaluation must always be performed by a certified medical specialist.